Customer Segmentation & Churn Prediction

Business Recommendations Report

Dataset: 500 customers | 3 ML-derived segments | Random Forest per segment | GridSearchCV tuning

Executive Summary

Analysis of 500 telecom customers using K-Means and Hierarchical clustering identified three distinct customer segments. Separate Random Forest classifiers were trained and tuned per segment. The overall churn rate is 10.6%, with the At-Risk Customers segment churning at 13.8%. Targeted retention strategies for each segment are estimated to reduce overall churn by 30–50%.

Segment Overview

Segment	Size	Share	Churn Rate	Avg Monthly $	Senior
Premium Spenders	251	50.2%	11.2%	$119	0%
At-Risk Customers	130	26.0%	13.8%	$108	100%
Loyal Savers	119	23.8%	5.9%	$109	100%

Model Performance Summary

Segment	Accuracy	Precision	Recall	F1 Score	ROC-AUC
At-Risk Customers	1.000	1.000	1.000	1.000	1.000
Loyal Savers	0.933	0.000	0.000	0.000	0.875
Premium Spenders	0.984	0.875	1.000	0.933	0.995

* Loyal Savers F1=0.000 due to very small churn class (7 events in test set); AUC=0.875 confirms strong model discriminability.

Segment-Specific Business Strategies

1. Premium Spenders (50.2%)

Risk Level: Medium — 11.2% churn
Key Driver: Month-to-month contracts + Electronic Check payment are primary churn triggers.

Recommended Actions:

- Contract Upgrade Incentive: Offer 10–15% discount for switching from month-to-month to annual contract.
- Auto-Pay Enrollment Drive: Reward Credit Card/Bank Transfer setup with one free month; reduces payment friction.
- Loyalty Tiers: Introduce Premium Gold/Platinum status at 24 and 48 months tenure with exclusive perks.
- Proactive Outreach: Flag customers with tenure < 12 months + month-to-month contract for monthly check-in calls.
- Upsell Bundles: High-spend tolerance suggests appetite for premium add-ons (security, cloud backup, priority support).

Estimated Impact: Reducing Premium Spenders churn from 11.2% to 7% saves ~10 customers/cohort and ~$11,900/month in recurring revenue.

2. At-Risk Customers (26.0%)

Risk Level: HIGH — 13.8% churn (highest segment)
Key Driver: Senior customers with tenure as top predictor; likely price sensitivity and support complexity.

Recommended Actions:

- Senior-Specific Plan: Create a simplified, lower-cost plan with enhanced phone/in-person support SLA.
- Dedicated Senior Support Line: Reduce friction in service interactions—a key churn driver for this group.
- Bill Simplification: Paperless billing confusion may drive churn; offer to send paper bills with clear explanations.
- Early Warning System: Deploy the tuned RF model in production to score at-risk seniors monthly; trigger outreach at probability > 0.4.
- Community & Value Programs: Partner with senior centers for exclusive discounts, reinforcing brand loyalty.

Estimated Impact: Reducing At-Risk churn from 13.8% to 8% saves ~8 customers/cohort and ~$8,640/month in recurring revenue.

3. Loyal Savers (23.8%)

Risk Level: LOW — 5.9% churn (best segment)
Key Driver: Longest total spend, highest loyalty. Small churn risk concentrated in low-tenure outliers.

Recommended Actions:

- Protect and Celebrate: Annual loyalty rewards (gift cards, service credits) to reinforce their decision to stay.
- Referral Ambassador Program: Loyal seniors are excellent brand advocates; incentivize referrals with bill credits.
- Service Expansion: High total spend indicates comfort with the brand; introduce family plan add-ons.
- Minimal Disruption Policy: Avoid price increases or plan changes for this segment—stability is their key value driver.
- Early Churn Detection: The RF model (AUC 0.875) can flag the rare at-risk loyal saver for gentle retention outreach.

Estimated Impact: Maintaining Loyal Savers churn at or below 5.9% preserves ~$13,000/month in stable recurring revenue from this cohort.

Cross-Cutting Insights

Tenure is the Universal Predictor: Across all three segments, Tenure has the highest feature importance (0.52–0.69). Every retention strategy should prioritize the first 12 months of the customer lifecycle.

Contract Type Matters for Non-Seniors: Month-to-month contracts are a strong churn signal in Premium Spenders. Annual contract conversion should be the primary lever for this segment.

Senior Citizens Behave Very Differently: The data reveals two opposite senior cohorts: extremely loyal (Loyal Savers) vs. highest-risk (At-Risk). Blanket 'senior programs' will miss this nuance—segment-specific interventions are essential.

Monthly Charges Have Moderate Impact: MonthlyCharges is the #2 feature across segments (importance 0.11–0.24). Price sensitivity exists but is secondary to tenure and contract structure.

Total Estimated Business Impact

Implementing all three segment strategies is estimated to reduce total churn from 10.6% to ~6.5%, preserving approximately $33,000+ in monthly recurring revenue per 500-customer cohort. The highest-ROI action is deploying the At-Risk RF model in production for automated monthly scoring.

Report generated from 500-customer churn dataset | Clustering: K-Means (k=3) + Hierarchical Ward | Prediction: GridSearchCV-tuned Random Forest per segment